

Elie Gambache

I build the **native desktop stack** the **JVM was missing**.

Creator of **Nucleus** — the native desktop platform for the JVM — and **30+ open-source libraries** for the Compose Multiplatform ecosystem. I wrap raw OS APIs in clean Kotlin and bridge Rust, C++ & Go into the JVM, so other developers don't have to.

◆ Jerusalem, Israel

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eliegambache.kdroidfilter.com

github.com/kdroidFilter

nucleusframework.dev

2,938GitHub stars
personal + NucleusFramework org**#6**GitHub committer
in Israel · committers.top**280K+**app downloads
shipping on my libraries

TRUSTED BY

JetBrains

ETH Zürich

Pushscroll 280K+

AB Download Manager 12K★

Sefaria

FLAGSHIP WORK

Nucleus [↗] — **FRAMEWORK**

★ 304 · Maven Central

The native desktop platform for the JVM

Gradle-plugin framework turning Compose Multiplatform into a full desktop stack — native windows, OS integration, code signing, installers — from one DSL. GraalVM native image (~0.5 s cold start, 60 MB RAM) or JVM. Push a tag → signed installers in 16 formats.

Gradle Plugin

GraalVM · JVM

FFM · JNI

Tray · Notifications · Hotkeys

Auto-Update

ComposeMediaPlayer [↗]

★ 475

The only Compose player on native codecs — no VLC, no MPV

Talks directly to MediaFoundation, AVPlayer, GStreamer, ExoPlayer & the browser — one @Composable, zero external dependencies. Supported by **JetBrains** · used by **Pushscroll** (280K+).

ComposeNativeTray [↗]

★ 398

A true native system tray — not AWT, not emulated

Declarative Compose API over the OS tray, each platform on its native backend: Go + D-Bus, C++, native macOS. Sponsored by **ETH Zürich** · used by **AB Download Manager** (12K★).

SELECTED OPEN-SOURCE LIBRARIES

ComposeNativeWebView ★ 162

OS-native WebView (WKWebView / WebView2 / WebKitGTK) via Wry + UniFFI — no Chromium, 90% smaller binaries.

ComposePdfReader ★ 59

KMP PDF rendering & text extraction on pdfium — Compose-first, progressive rendering, selectable text.

DesktopAccessibilityManager ETH Zürich

announceForAccessibility() for Compose Desktop via JNA — bridges NVDA, JAWS, VoiceOver.

AudioKt Rust

JVM audio engine wrapping Rust's Rodio — local files, HLS, internet radio with ICY metadata.

APPLICATIONS

Zayit

4,200+ downloads

Semantic search over 7,000+ Jewish texts – fully offline

Production KMP app: Hebrew-aware tokenization, full-text indexing, citation-aware navigation on SQLite. Powered by Sefaria.

CORE STACK

LANGUAGES

Kotlin · Rust · C++ · Go · Python · TypeScript

FRAMEWORKS

Compose Multiplatform · KMP · React

NATIVE BRIDGING

JNI · Java FFM · JNA · UniFFI · GraalVM native image

PLATFORM APIS

Win32 / WinRT · AppKit / AVFoundation · D-Bus · MediaFoundation · GStreamer

BUILD & DELIVERY

Gradle plugins · GitHub Actions · signing · notarization · installers · auto-update

DOMAINS

Android MDM & enterprise · search engines · Hebrew NLP

FREELANCE & CONSULTING

Systems architecture

KMP library dev

Compose Desktop apps

Native integration

Gradle plugins

Android MDM

Search & data pipelines

Architecture review